

Supporting Information

A Differentiable Ray–Wave Framework for Hybrid Refractive–Diffractive System Modeling and Optimization

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S1 Existing Ray–wave Framework Comparison

Table S1: Comparison of ray–wave frameworks across key capabilities for hybrid optical systems.

Method	Conformal DOE	Holographic Phase DOE	Complex- Amplitude Modulation	Flexible DOE Placement	Non- paraxial	Differentiable
Yu et al. ¹	✓	✗	✗ (phase only)	✓	✓	✓
Mout et al. ²	✗	✗	✗ (amp. only)	✓	✓	✗
Zhu et al. ³	✓	✗	✗ (phase only)	✓	✓	✓
Shih et al. ⁴	✗	✗	✓	✓	✓	✓
Yang et al. ⁵	✗	✓	✓	✗ (last)	✓	✓
Zhang et al. ⁶	✓	✗	✗ (phase only)	✓	✓	✓
Cheng et al. ⁷	✗	✗	✓	✓	✓	✗
Shi et al. ⁸	✓	✗	✗ (phase only)	✓	✓	✓
Ellepola et al. ⁹	✓	✗	✗ (amp. only)	✓	✓	✗
Our work	✓	✓	✓	✓	✓	✓

Table S1 summarizes the scope of representative ray–wave frameworks. Compared with existing ray–wave approaches, our method supports planar and conformal DOEs, holographic phase and complex-amplitude modulation, flexible DOE placement within a multi-element optical system, non-paraxial propagation, and differentiable optimization. The primary contribution is therefore not universal computational superiority, but a differentiable DOE module that combines wave-optical scattering with conventional ray tracing for coherent modeling and inverse design of a broad range of hybrid refractive–diffractive systems.

For free-space propagation between planar surfaces, direct scalar wave-propagation methods, such as ASM, provide deterministic and often more computationally efficient solutions. Likewise, for DOEs with locally smooth phase profiles, such as gratings or low-NA metalenses, phase-gradient ray tracing based on the generalized law of refraction can be sufficient. Our method provides a useful reference for identifying when the local smooth-phase approximation begins to fail.

S2 Secondary-Ray Sampling and Planar Patch Representation

The key claim is that each local diffractive field patch $U(u, v)$ at a ray–DOE intersection is propagated exactly by the angular spectrum method (ASM), and that our secondary ray generation is an unbiased Monte Carlo estimator of that per-patch ASM integral.

Consider the complex-amplitude patch $U(u, v)$ extracted at the ray–DOE intersection in the local tangent frame $(\hat{\mathbf{u}}, \hat{\mathbf{v}}, \hat{\mathbf{n}})$. Its Fourier transform

$$\tilde{U}(k_u, k_v) = \iint U(u, v) e^{-i(k_u u + k_v v)} du dv \quad (\text{S1})$$

defines the local angular spectrum, where (k_u, k_v) are transverse wavevectors in the tangent plane. Components satisfying $k_u^2 + k_v^2 > k^2$ are evanescent and are discarded; only propagating modes with $k_n = \sqrt{k^2 - k_u^2 - k_v^2} \in \mathbb{R}$ are retained. The ASM gives the exact diffracted field at a downstream observation point $\mathbf{r} = (x, y, d)$ as

$$U_{\text{sensor}}(\mathbf{r}) = \iint \tilde{U}(k_u, k_v) e^{i(k_u x + k_v y + k_n d)} dk_u dk_v. \quad (\text{S2})$$

Rather than evaluating the continuous ASM integral directly, we approximate it by drawing N spectral-component samples $\{(k_u^{(i)}, k_v^{(i)})\}_{i=1}^N$, with each sampled component corresponding to a plane wavelet with wavevector $\mathbf{k}^{(i)} = (k_u^{(i)}, k_v^{(i)}, k_n^{(i)})$. We associate this wavelet with a secondary ray whose propagation direction is $\mathbf{d}^{(i)} = \mathbf{k}^{(i)}/k$, and replace the integral with a discrete summation over secondary rays,

$$U_{\text{sensor}}(\mathbf{r}) \approx \frac{1}{N} \sum_{i=1}^N \tilde{U}(k_u^{(i)}, k_v^{(i)}) e^{i(k_u^{(i)} x + k_v^{(i)} y + k_n^{(i)} d)}. \quad (\text{S3})$$

Each sampled spectral component generates a secondary ray whose contribution to the sensor field consists of two parts:

- **Complex amplitude:** Each sampled spectral component assigns a complex amplitude to its corresponding secondary ray, $a^{(i)} = \tilde{U}(k_u^{(i)}, k_v^{(i)})$.
- **Optical path length:** For free-space propagation from the DOE to a plane located at axial distance d , the secondary ray accumulates optical path length $\text{OPL}^{(i)} = \mathbf{d}^{(i)} \cdot \mathbf{r} = \frac{1}{k} \mathbf{k}^{(i)} \cdot \mathbf{r}$, which produces the phase factor $e^{ik \text{OPL}^{(i)}} = e^{i(k_u^{(i)}x + k_v^{(i)}y + k_n^{(i)}d)}$ at the sensor.

The sensor field is therefore obtained by coherently summing the complex contributions of all secondary rays.

The sampling strategy of the secondary rays strongly affects the convergence of this stochastic coherent summation. In general, the spectral components are sampled from a probability density function $p(k_u, k_v)$ defined over the angular spectrum domain. To maintain an unbiased estimate under this non-uniform sampling,¹⁰ the complex amplitude carried by each secondary ray is rescaled according to the sampling density,

$$a^{(i)} = \frac{\tilde{U}(k_u^{(i)}, k_v^{(i)})}{p(k_u^{(i)}, k_v^{(i)})}. \quad (\text{S4})$$

The sensor field for free-space propagation along direction $\hat{\mathbf{n}}$ is therefore computed as the coherent summation of secondary rays,

$$U_{\text{sensor}}(\mathbf{r}) \approx \frac{1}{N} \sum_{i=1}^N a^{(i)} e^{ik \text{OPL}^{(i)}}. \quad (\text{S5})$$

Because the formulation expresses the field as a sum of wavelets carried by rays, it naturally integrates with conventional ray tracing frameworks, where rays are already interpreted as wavelets and coherent fields are computed through Huygens PSF calculation. In this sense, our ray-DOE formulation can be seamlessly embedded within existing coherent ray-tracing pipelines.

Physics Intuition Behind Patch Representation for Planar DOEs. For planar DOEs, there is no intrinsic upper bound on the patch size because the tangent-plane approx-

imation is exact everywhere on the surface. As long as the ensemble of patches uniformly covers the DOE profile, the coherent sum of their responses converges to the full DOE response through coherent superposition, consistent with the linearity of the Fourier transform. Increasing the patch size only changes how the DOE response is partitioned among primary rays. A larger patch captures a broader spatial portion of the DOE response and correspondingly reduces the number of incident rays needed for full-aperture coverage.

This behavior can also be understood from the full-field-patch picture (main text Figure 2) in Figure S1. When the entire planar DOE is treated as a single patch, one planar-wave mode can be represented by a collimated bundle of rays with the same propagation direction but different lateral origins (Figure S1c). The rays are assigned relative phases that compensate for the OPL differences associated with their lateral positions, thereby preserving the same plane-wave mode across the full aperture.

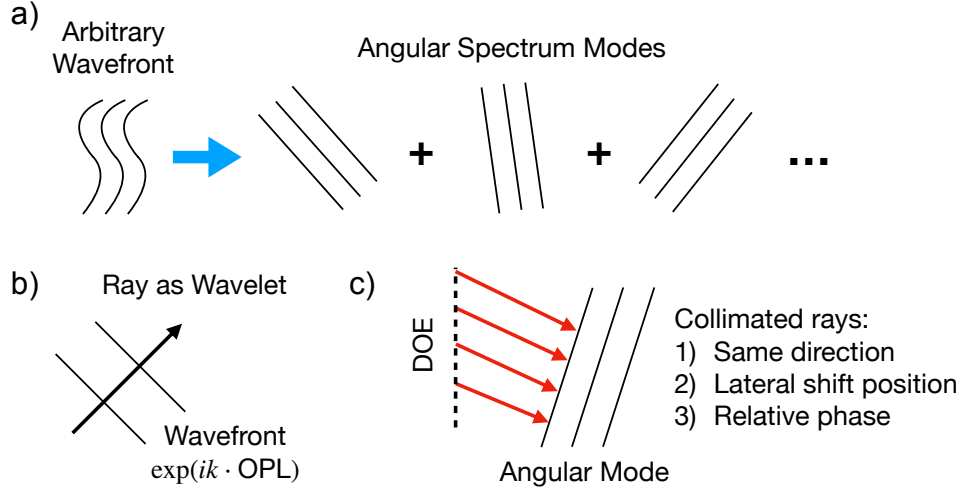


Figure S1: Physical interpretation of ray-wave modeling. (a) An arbitrary wavefront can be decomposed into a superposition of plane-wave modes with different propagation wavevector, based on the angular spectrum method. (b) In conventional ray-tracing representation, each ray can be treated as a plane-wave wavelet that carries both a propagation direction and a phase determined by its optical path length (OPL), $\exp(ik \cdot \text{OPL})$. By recording the phase change, we can track the planar wavefront of the ray. (c) Physics intuition for the full-field-patch representation. A collimated ray bundle with rays launched from multiple lateral positions on a DOE can represent a single angular-spectrum mode. Although all rays share the same propagation direction (wavevector), rays starting from different lateral positions must travel different OPLs to reach the same planar wavefront. Their phases thus have offsets that compensate for these OPL differences, preserving the phase relation of the corresponding plane-wave mode.

S3 Derivation of Curvature-induced Error Bound

We first analyze the effect of surface curvature on the propagation direction extracted from the local angular spectrum in one dimension along a principal curvature direction; the result generalizes directly to two dimensions by applying the same argument along both principal curvature axes. Consider a diffractive surface with local radius of curvature (RoC) R . Over a finite patch of width D centered at the origin, the surface sag introduces an additional quadratic phase,

$$\phi_{\text{curv}}(x) = \frac{1}{2} \frac{k_0}{R} x^2, \quad (\text{S6})$$

where $k_0 = 2\pi/\lambda$ is the free-space wavenumber. The local spatial frequency is obtained from the phase gradient,

$$f(x) = \frac{1}{2\pi} \frac{d\phi_{\text{curv}}}{dx} = \frac{k_0}{2\pi R} x. \quad (\text{S7})$$

This shows that the additional spatial frequency varies linearly with position across the region and thus produces a broadened angular spectrum.

The maximum deviation of spatial frequency occurs at the region edges $x = \pm D/2$, which gives a bound on the frequency spread,

$$|\Delta f| \leq \frac{k_0}{2\pi R} \frac{D}{2}. \quad (\text{S8})$$

This frequency spread translates into an angular spread in the sampled propagation directions. Since propagation angle is related to spatial frequency through $\sin \theta = \lambda f$, the curvature-induced direction error is bounded by

$$\varepsilon_{\text{curv}} \leq \arcsin \left(\lambda \cdot \frac{k_0}{2\pi R} \frac{D}{2} \right) = \arcsin \left(\frac{D}{2R} \right). \quad (\text{S9})$$

S4 Field Accumulation and Ray Resampling for Cascaded Planar DOEs

Algorithm S1 Ray Accumulation and Resampling for Cascaded Planar DOE

Require: Incoming ray bundle $\mathcal{R}_{\text{in}}$, planar DOE profile $U_{\text{DOE}}(x, y)$, plane grid Ω , wavelength λ , output primary-ray count P , secondary-ray count per primary ray S

Ensure: Resampled outgoing ray bundle $\mathcal{R}_{\text{out}} = \left\{ \left(\mathbf{o}^{(ps)}, \mathbf{d}^{(ps)}, a^{(ps)}, \text{OPL}^{(ps)} \right) \right\}_{p=1, s=1}^{P, S}$

```

1:  $U_{\text{in}}(x, y) \leftarrow \text{RAYTOFIELD}(\mathcal{R}_{\text{in}}, \Omega)$  ▷ coherently sum all incident rays on the plane
2:  $U_{\text{out}}(x, y) \leftarrow U_{\text{in}}(x, y) U_{\text{DOE}}(x, y)$  ▷ apply complex DOE profile
3:  $\tilde{U}_{\text{out}}(k_x, k_y) \leftarrow \mathcal{F}\{U_{\text{out}}(x, y)\}$  ▷ angular spectrum
4:  $p(k_x, k_y) \leftarrow f(\tilde{U}_{\text{out}}(k_x, k_y))$  ▷ PDF over propagating spectral modes
5:  $\{(x^{(p)}, y^{(p)})\}_{p=1}^P \leftarrow \text{SAMPLEPLANEPOSITIONS}(\Omega)$  ▷ choose outgoing ray origins on the DOE plane
6: for  $p = 1, \dots, P$  do
7:   for  $s = 1, \dots, S$  do
8:      $(k_x^{(ps)}, k_y^{(ps)}) \sim p$  ▷ sample outgoing wavevector
9:      $\phi^{(ps)} \leftarrow k_x^{(ps)} x_p + k_y^{(ps)} y_p$  ▷ relative phase for ray origin
10:     $\mathbf{o}^{(ps)} \leftarrow (x^{(p)}, y^{(p)}, z_{\text{DOE}})$  ▷ ray launching position
11:     $\mathbf{d}^{(ps)} \leftarrow \left( k_x^{(ps)}, k_y^{(ps)}, \sqrt{k^2 - k_x^{(ps)2} - k_y^{(ps)2}} \right) / k$  ▷ ray direction
12:     $\text{OPL}^{(ps)} \leftarrow 0$  ▷ ray initialized optical path length
13:     $a^{(ps)} \leftarrow \frac{\tilde{U}_{\text{out}}(k_x^{(ps)}, k_y^{(ps)})}{p(k_x^{(ps)}, k_y^{(ps)})} e^{i\phi^{(ps)}}$  ▷ ray complex amplitude
14:   end for
15: end for
16: return  $\mathcal{R}_{\text{out}}$ 

```

For concatenated planar DOEs, the direct implementation of our framework can propagate the sampled secondary rays from the previous DOE to the next DOE, where these rays become the incident-ray ensemble for the next DOE interaction. While this brute-force procedure is valid, repeatedly sampling secondary rays at successive planar DOE interactions can cause the ray count to grow exponentially.

A more efficient alternative (Algorithm S1) is available for planar DOE surfaces. Because

all rays intersect a common Cartesian plane, the complete incoming ray bundle can first be coherently accumulated into a complex field U_{in} on that plane, in the same manner as the field reconstruction at the sensor plane. The DOE transmission is then applied once to this accumulated field to get the output field U_{out} , after which a single global FFT computes the outgoing angular spectrum. A fixed number of outgoing rays, including P primary ray locations and S secondary rays per primary location, is subsequently sampled from this spectrum using Monte Carlo sampling. Therefore, the ray count after a planar DOE is controlled by a prescribed output budget rather than by the product of the incoming-ray count and the number of sampled secondary rays. This operation preserves coherent interference at the planar DOE while preventing ray-number growth across cascaded planar interactions.

For conformal DOEs, this global aggregation before ray–DOE interaction is not applicable because rays intersect different local tangent planes with position-dependent coordinate frames and surface normals. We therefore retain the direct implementation.

S5 Peak GPU Memory Usage for the Sampling Sweep in Figure 3b

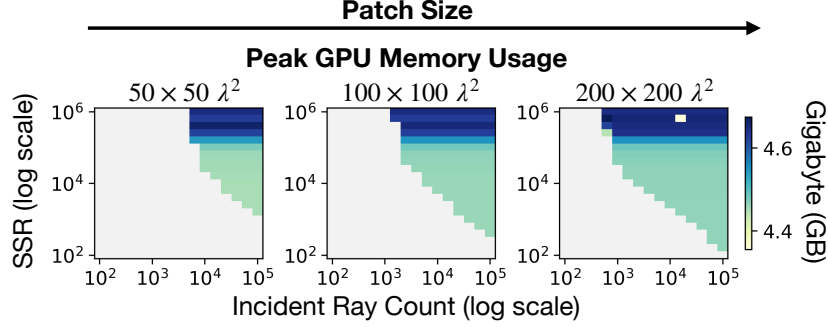


Figure S2: Peak GPU memory usage for the ray-count sweep in Figure 3b across different DOE patch sizes. For each incident-ray and SSR configuration, the ray batch size was chosen as the largest value that could be processed without exceeding available GPU memory. Cells with mean NCC below 0.99 are omitted.

To ensure that the runtime measurements in Figure 3b are directly comparable across ray-count configurations, we first determined the largest ray batch that could be processed on the GPU for each combination of incident-ray count, SSR count, and DOE patch size. The ray batches were then processed sequentially, with the coherent sensor field accumulated across batches. This procedure maximizes GPU utilization while avoiding out-of-memory errors, such that the reported wall times reflect the computational cost of each converged configuration under near-maximal available GPU memory usage. For each configuration, we recorded the peak allocated GPU memory during the simulation and averaged the measurements across repeated trials. Figure S2 shows the resulting peak memory cost for the configurations in Figure 3b with mean NCC ≥ 0.99 .

S6 Monte Carlo Convergence in the Hologram–Lens System

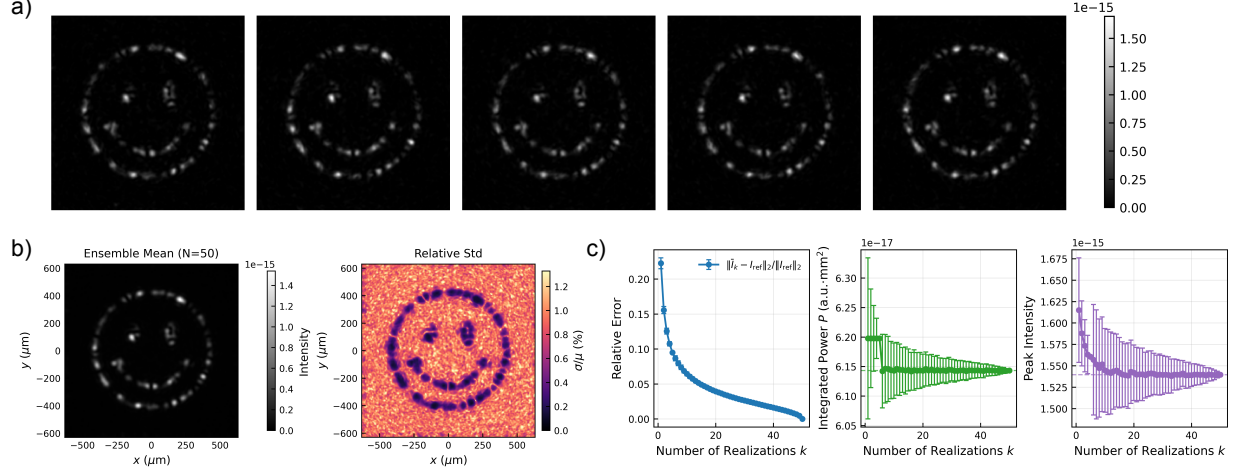


Figure S3: Monte Carlo ensemble convergence across $N = 50$ realizations of the hologram–lens system simulation with ray–wave modeling. (a) 5 independent stochastic realizations generated with different random seeds. (b) The ensemble averaged intensity I_{ref} over $N = 50$ realizations and the corresponding pixelwise relative standard deviation σ_I / μ_I , where σ_I is the pixel intensity standard deviation and μ_I is the pixel mean intensity. (c) Left: relative image error of the k -realization averaged intensity with respect to I_{ref} , $\|\bar{I}_k - I_{\text{ref}}\|_2 / \|I_{\text{ref}}\|_2$, where $\bar{I}_k$ denotes the mean intensity over a subset of k independently sampled realizations. Middle: integrated sensor power of $\bar{I}_k$. Right: peak intensity of $\bar{I}_k$. For each k , the marker denotes the mean metric evaluated over multiple subsets of k realizations, and the error bar denotes the standard deviation across these subsets, quantifying the dependence of the estimate on the selected random seeds. For $k \leq 5$, the subsets are obtained from all 5! orderings of the first five realizations; for $k > 5$, 500 subsets are randomly sampled from the full ensemble of 50 realizations. Dashed lines indicate the integrated power and peak intensity of I_{ref} . Increasing the number of realizations reduces both image-level error and realization-dependent fluctuations in the integrated power and peak intensity.

To assess the stochastic convergence of the ray–wave simulation, we performed $N = 50$ independent realizations with different random seeds. In Figure S3b, realization-to-realization variation is concentrated primarily in dim background regions. The relative standard deviation is correspondingly large where the mean intensity approaches zero, as expected for weak fields; nevertheless, the bright image features and the dominant speckle structure exhibit substantially smaller variation and remain spatially reproducible across independent

realizations. The ensemble-averaged intensity provides a stable reference image, and the relative error of the k -realization average decreases steadily as k increases, indicating convergence of the Monte Carlo estimate. The integrated sensor power and peak intensity also converge toward the corresponding values of the ensemble-averaged reference, with decreasing variation across realization subsets.

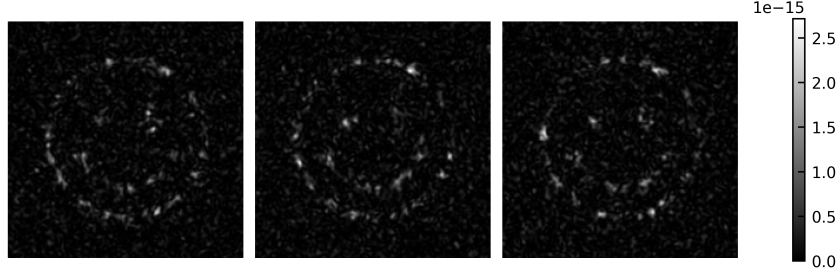


Figure S4: Artifacts caused by insufficient Monte Carlo convergence in the hologram–lens system. When the incident-ray and SSR counts are insufficient, stochastic interference produces strong speckle-like artifacts across the whole sensor plane.

To further isolate numerical artifacts, we intentionally evaluate the hologram–lens system using insufficient incident-ray and SSR counts. As shown in Figure S4, undersampling produces strong, seed-dependent speckle-like fluctuations across the entire sensor plane. These fluctuations differ qualitatively from the converged result, such that they are not spatially reproducible across realizations.

S7 Differentiable Ray–Wave Modeling and Gradient Estimation

We optimize the DOE profile using gradient-based optimization with auto-differentiation in PyTorch. The forward model consists of a stochastic ray–wave simulation in which the complex field at the sensor plane is obtained as a coherent sum of contributions from many sampled rays. Because the number of rays required for accurate simulation is very large, the rays are processed in batches rather than all at once. This batching strategy allows the simulation to fit within GPU memory while maintaining a large total ray count.

S7.1 Batched Forward Model

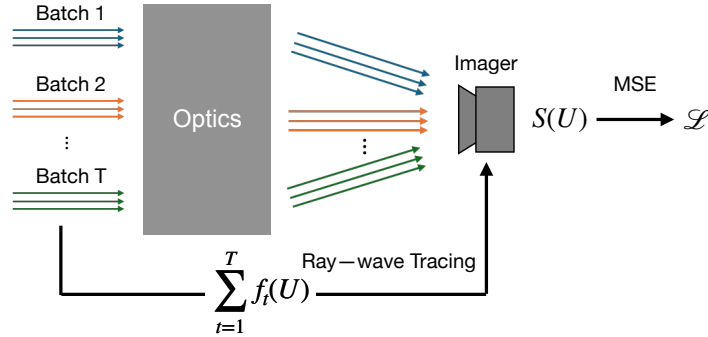


Figure S5: Forward simulation schematics of ray–wave framework with ray batches.

Let the complex field at the sensor plane be

$$S(U) = \sum_{t=1}^T f_t(U), \quad (\text{S10})$$

where U denotes the DOE profile and f_t represents the complex field contribution from the t -th batch of rays. Each batch contains many rays that are simulated simultaneously on the GPU, and the final field is obtained by coherently summing the contributions from all ray batches.

S7.2 Fixed-Direction Gradient Estimator

Algorithm S2 summarizes the gradient estimator used in our implementation. In the forward pass, each secondary-ray wavevector $(k_u^{(s)}, k_v^{(s)})$ is sampled discretely from the angular-spectrum probability distribution $p(k_u, k_v)$, and the corresponding outgoing direction $\mathbf{d}^{(s)}$ is computed from the selected wavevector. Because this discrete sampling operation is non-differentiable, the sampled wavevector and ray direction are treated as fixed during backpropagation.

Algorithm S2 Patched-Field Ray-Direction Sampling

Require: Patch $U(u, v)$, input direction $\mathbf{d}_{\text{in}}$, surface normal $\mathbf{n}$, wavenumber k , SSR count S

Ensure: $\{(\mathbf{d}^{(s)}, a^{(s)})\}_{s=1}^S$

- 1: $\tilde{U} \leftarrow \mathcal{F}\{U\}$ ▷ angular spectrum
 - 2: $p \leftarrow f(\tilde{U})$ ▷ PDF from spectrum
 - 3: **for** $s = 1, \dots, S$ **do**
 - 4: $(k_u^{(s)}, k_v^{(s)}) \sim p$ ▷ sample k-vector
 - 5: $\mathbf{d}^{(s)} \leftarrow \mathbf{d}_{\text{in}} + \left(k_u^{(s)}, k_v^{(s)}, \sqrt{k^2 - k_u^{(s)2} - k_v^{(s)2}} \right) / k$ ▷ outgoing direction (no grad)
 - 6: $a^{(s)} \leftarrow \frac{\tilde{U}(k_u^{(s)}, k_v^{(s)})}{p(k_u^{(s)}, k_v^{(s)}) \cdot \text{detach}()}$ ▷ ray amplitude (with grad)
 - 7: **end for**
 - 8: **return** $\{(\mathbf{d}^{(s)}, a^{(s)})\}_{s=1}^S$
-

S7.3 Gumbel–Softmax as an Alternative Differentiable Sampler

We consider a Gumbel–Softmax reparameterization method as an alternative approach for differentiating through discrete secondary-ray sampling. In the forward pass, the method is statistically equivalent to categorical sampling of wavevector bins from the angular-spectrum probability distribution $p(k_u, k_v)$. In the backward pass, it replaces the discrete wavevector selection with a softmax-based gradient estimate.

For a DOE patch, let $p(k_u, k_v)$ denote the normalized probability distribution over the discretized $qD \times qD$ angular-spectrum grid, where $D \times D$ is the unpadded patch size and q is the zero-padding factor. We flatten the propagating wavevector grid into discrete bins

indexed by j , where each bin corresponds to a transverse wavevector pair $(k_{u,j}, k_{v,j})$ and has probability p_j . For each secondary ray s , we draw an independent Gumbel-noise vector $\mathbf{g}^{(s)}$. Its j th entry is generated by first sampling $u_j^{(s)} \sim \mathcal{U}(0, 1)$ and then computing

$$g_j^{(s)} = -\log\left[-\log\left(u_j^{(s)}\right)\right]. \quad (\text{S11})$$

We add this noise vector to the log-probabilities (logits) $\ell_j = \log p_j$. By the Gumbel-max identity,¹¹ selecting the bin with the largest perturbed logit is statistically equivalent to sampling directly from $p(k_u, k_v)$:

$$\Pr\left[\text{onehot}\left(\arg\max_j \left[\ell_j + g_j^{(s)}\right]\right) = \mathbf{e}_j\right] = p_j, \quad (\text{S12})$$

where $\mathbf{e}_j$ is the one-hot vector corresponding to wavevector bin j . The hard selected wavevector bin is therefore

$$\mathbf{y}_{\text{hard}}^{(s)} = \text{onehot}\left(\arg\max_j \left[\ell_j + g_j^{(s)}\right]\right). \quad (\text{S13})$$

Thus, the Gumbel-max procedure preserves the original wavevector sampling distribution. The forward Monte Carlo field reconstruction is unchanged and introduces no additional bias relative to direct categorical sampling from $p(k_u, k_v)$.

Although the Gumbel-max procedure preserves the correct forward sampling distribution, the resulting hard one-hot vector $\mathbf{y}_{\text{hard}}^{(s)}$ is still not differentiable with respect to the sampling probabilities p . To obtain gradients, we compute the continuous relaxed counterpart of $\mathbf{y}_{\text{hard}}^{(s)}$ with the softmax formulation

$$\mathbf{y}_{\text{soft}}^{(s)} = \text{softmax}\left(\frac{\ell + \mathbf{g}^{(s)}}{\tau}\right), \quad (\text{S14})$$

where τ is a temperature parameter that controls the sharpness of the relaxation. We then construct

$$\bar{\mathbf{y}}^{(s)} = \mathbf{y}_{\text{hard}}^{(s)} + \mathbf{y}_{\text{soft}}^{(s)} - \mathbf{y}_{\text{soft}}^{(s)}. \text{detach}(). \quad (\text{S15})$$

Consequently, this construction preserves the hard selected wavevector bin $\mathbf{y}_{\text{hard}}^{(s)}$ in the forward pass, while allowing gradients to be evaluated through the continuous softmax counterpart $\mathbf{y}_{\text{soft}}^{(s)}$ during backpropagation.¹² Thus, the Gumbel–Softmax approximation affects only the gradient calculation and does not alter the forward wavevector sampling statistics or the forward field reconstruction relative to direct categorical sampling. The resulting backpropagated gradient is an approximate, generally biased surrogate for the gradient of the discrete sampling operation. Algorithm S3 summarizes the complete procedure.

The quality of this gradient approximation is governed by the temperature parameter τ , which controls the sharpness of the gradient relaxation (Figure S6). Smaller values make $\mathbf{y}^{(s)}$ closer to the hard selected bin but can yield unstable gradients, whereas larger values provide smoother gradients but a less accurate approximation. We choose $\tau = 3$ heuristically for our ray–wave modeling, since this value provided sufficiently smooth optimization curves and consistently converged to reasonable reconstructed solutions.

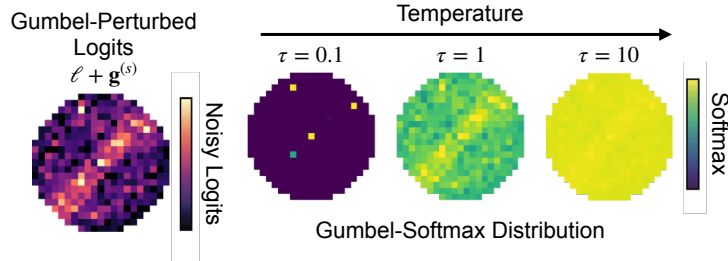


Figure S6: Effect of the Gumbel–Softmax temperature on the continuous relaxation of discrete wavevector sampling. Left: Gumbel-perturbed logits, $\ell + \mathbf{g}^{(s)}$, over the discretized angular-spectrum grid for one sampled secondary ray s . Right: the corresponding Gumbel–Softmax distributions evaluated at different temperatures. A small temperature ($\tau = 0.1$) produces a nearly one-hot distribution concentrated at the maximum-logit wavevector bin, whereas increasing τ yields progressively smoother and more spatially distributed probability weights.

Although this method provides a direct differentiable relaxation of categorical secondary-ray sampling, we did not use it in the final implementation because of its memory cost. For a batch containing N_t rays, corresponding to the product of the incident-ray count and the SSR count, the method must retain a Gumbel–Softmax tensor for every ray over the full padded

angular-spectrum grid. The resulting intermediate tensor has dimensions $N_t \times qD \times qD$. Because both N_t and qD must be large for accurate Monte Carlo ray-wave simulations, this storage becomes prohibitively large on GPU memory. Therefore, we use the gradient estimator described in the main text and Section 7.2, in which sampled ray directions are held fixed during backpropagation and gradients propagate only through the complex amplitudes carried by the sampled rays.

Algorithm S3 Patched-Field Ray-Direction Sampling with Gumbel-Softmax

Require: Patch $U(u, v)$, input direction $\mathbf{d}_{\text{in}}$, surface normal $\mathbf{n}$, wavenumber k , SSR count S , temperature τ

Ensure: $\{(\mathbf{d}^{(s)}, a^{(s)})\}_{s=1}^S$

- 1: $\tilde{U} \leftarrow \mathcal{F}\{U\}$ ▷ angular spectrum
 - 2: $p \leftarrow f(\tilde{U})$ ▷ PDF from spectrum
 - 3: $\ell \leftarrow \log(p)$ ▷ Gumbel logits
 - 4: **for** $s = 1, \dots, S$ **do**
 - 5: $\mathbf{g}^{(s)} \leftarrow -\log(-\log(\mathbf{u}^{(s)})), \quad \mathbf{u}^{(s)} \sim \mathcal{U}(0, 1)$ ▷ Gumbel noise
 - 6: $\mathbf{y}_{\text{soft}}^{(s)} \leftarrow \text{softmax}((\ell + \mathbf{g}^{(s)})/\tau)$ ▷ Gumbel-Softmax
 - 7: $\mathbf{y}_{\text{hard}}^{(s)} \leftarrow \text{onehot}(\arg \max(\ell + \mathbf{g}^{(s)}))$
 - 8: $\bar{\mathbf{y}}^{(s)} = \mathbf{y}_{\text{hard}}^{(s)} + \mathbf{y}_{\text{soft}}^{(s)} - \mathbf{y}_{\text{soft}}^{(s)}.\text{detach}()$ ▷ forward hardmax, backward softmax
 - 9: $(k_u^{(s)}, k_v^{(s)}) \leftarrow \mathcal{K}(\bar{\mathbf{y}}^{(s)})$ ▷ select k-vector
 - 10: $\mathbf{d}^{(s)} \leftarrow \mathbf{d}_{\text{in}} + \left(k_u^{(s)}, k_v^{(s)}, \sqrt{k^2 - k_u^{(s)2} - k_v^{(s)2}} \right) / k$ ▷ outgoing direction (with grad)
 - 11: $a^{(s)} \leftarrow \frac{\tilde{U}(k_u^{(s)}, k_v^{(s)})}{p(k_u^{(s)}, k_v^{(s)})}$ ▷ ray amplitude (with grad)
 - 12: **end for**
 - 13: **return** $\{(\mathbf{d}^{(s)}, a^{(s)})\}_{s=1}^S$
-

S8 Computational Cost and Scalability

To provide practical guidance for readers and future users, we summarize the computational specs, memory usage, and runtime of the ray-wave framework used in benchmark and inverse design. All simulations were performed on a single NVIDIA RTX A6000 GPU with 48 GB of memory using CUDA 12.4. All benchmark and inverse-design demonstrations were performed on a single GPU without multi-GPU parallelization.

All benchmark simulations are implemented in PyTorch by extending the differentiable ray tracer DeepLens¹³ with our ray-wave DOE surface module. The reported runtime and memory usage therefore include both the conventional ray tracer from DeepLens and a plug-and-play ray-wave module introduced by our method, including DOE patch extraction, angular spectrum calculation, secondary ray sampling, and coherent sensor-field reconstruction. For the grating-lens system, the ground-truth (GT) result is computed using a generalized-law-of-refraction diffractive surface implemented within the same DeepLens framework. For the free-space hologram system, the GT result is computed using the angular spectrum method (ASM).

Table S2: Computational cost of benchmark simulations with open-source ray tracer DeepLens.¹³ “RW-P” denotes the local patch-based implementation, whereas “RW-F” denotes the planar DOE implementation that uses one global zero-padded angular spectrum.

System	DOE (px ²)	Patch (px ²)	Pad factor	Incident rays	SSR	Ray Batches	Peak Mem. (GB)	Runtime (sec)
Grating-Lens (GT)	4032 ²	/	/	1.6×10^6	/	/	0.386	0.045
Grating-Lens (RW-P)	4032 ²	40 ²	4	6.5×10^4	10 ³	8	11.492	4.982
Grating-Lens (RW-F)	4032 ²	Full	/	/	/	/	OOM	/
Hologram (GT)	100 ²	/	/	/	/	/	0.003	0.040
Hologram (RW-P)	100 ²	50 ²	4	1.6×10^4	10 ⁴	2	29.213	2.275
Hologram (RW-F)	100 ²	Full	/	1	1.1×10^6	1	8.086	0.097
DOE-Lens (RW-P)	200 ²	100 ²	2	2.6×10^5	10 ⁴	8	17.003	41.137
DOE-Lens (RW-F)	200 ²	Full	/	3.3×10^4	1.6×10^5	128	16.919	28.387

For inverse-design experiments, we use a customized lightweight implementation with the same ray-wave operations as the DeepLens plug-in. Please note that the DOE-lens inverse-design simulation uses 256 incident rays and 8.2×10^3 secondary rays per iteration

as a stochastic minibatch to reduce the cost of repeated forward and backward evaluations, whereas the benchmark simulation uses a substantially larger ray ensemble, 3.3×10^4 incident rays and 1.6×10^5 secondary rays, to obtain a more converged reconstruction of the sensor field.

Table S3: Computational cost of inverse-design demonstrations. “RW-P” denotes the local patch-based ray-wave implementation, whereas “RW-F” denotes the planar DOE implementation that uses one global zero-padded angular spectrum. The DOE-lens optimization corresponds to the hybrid refractive-diffractive system in Figure 6. The beamsplitter and Stanford “S” optimizations correspond to the conformal reflective DOE demonstrations in Figure 7.

Task	Ray-wave Model	Patch (px ²)	Pad factor	Incident rays	Secondary rays	Epochs /Batches	Peak Mem. (GB)	Runtime (min)
DOE-Lens	RW-F	100 ²	/	256	8.2×10^3	100/128	0.356	13.587
Beamsplitter	RW-P	201 ²	2	3.0×10^3	4.0×10^3	50/1	10.017	4.752
Stanford “S”	RW-P	201 ²	2	3.0×10^3	4.0×10^3	50/2	38.440	5.849

S9 Inverse-Design Details and Additional Results

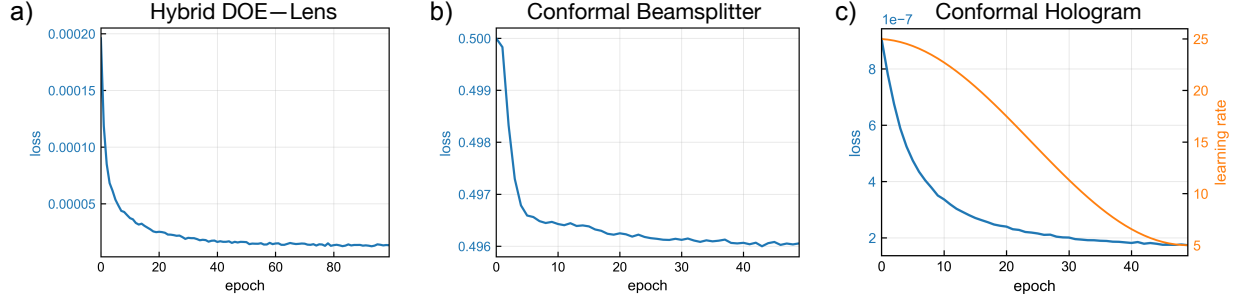


Figure S7: Loss trajectories for the inverse-design demonstrations. (a) Hologram–lens optimization in Figure 6 using a fixed learning rate of 10.0. (b) Conformal beamsplitter optimization in Figure 7c,d using a fixed learning rate of 1.0. (c) Conformal hologram optimization in Figure 7e,f using a scheduled learning rate.

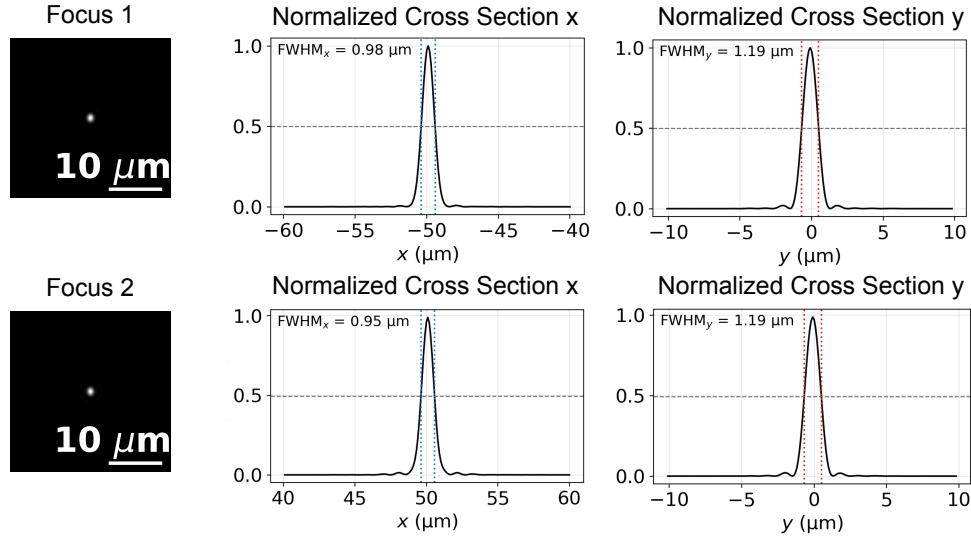


Figure S8: Normalized cross sections of the two optimized focusing spots along the x - and y -axes for the conformal beamsplitter result in Figure 7d. Top row: focus 1. Bottom row: focus 2.

References

- (1) Yu, N.; Genevet, P.; Kats, M. A.; Aieta, F.; Tetienne, J.-P.; Capasso, F.; Gaburro, Z. Light Propagation with Phase Discontinuities: Generalized Laws of Reflection and Refraction. *Science* **2011**, *334*, 333–337.
- (2) Mout, M.; Flesch, A.; Wick, M.; Bociort, F.; Petschulat, J.; Urbach, P. Ray-based method for simulating cascaded diffraction in high-numerical-aperture systems. *Journal of the Optical Society of America A* **2018**, *35*, 1356.
- (3) Zhu, Z.; Liu, Z.; Zheng, C. Metalens enhanced ray optics: an end-to-end wave-ray co-optimization framework. *Optics Express* **2023**, *31*, 26054–26068.
- (4) Shih, K.-H.; Renshaw, C. K. Hybrid meta/refractive lens design with an inverse design using physical optics. *Applied Optics* **2024**, *63*, 4032–4043.
- (5) Yang, X.; Souza, M.; Wang, K.; Chakravarthula, P.; Fu, Q.; Heidrich, W. End-to-End Hybrid Refractive-Diffractive Lens Design with Differentiable Ray-Wave Model. SIGGRAPH Asia 2024 Conference Papers. Tokyo Japan, 2024; pp 1–11.
- (6) Zhang, Q.; Lin, P.; Yu, Z.; Zhang, C.; Liu, Y.; Wang, M.; Fan, Q.; Wang, C.; Xu, T.; Zheng, Z. Vectorial Generalized Snell's Law-Enabled Differentiable Ray Tracing for Large-Aperture Visible Achromatic Hybrid Meta-Optics. *Laser & Photonics Reviews* **2025**, *19*, e00448, reprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/lpor.202500448>.
- (7) Cheng, H.-H. M.; Leportier, T.; Huynh, D.-N.; Niegemann, J.; Reid, A.; Chen, W.-H. Ray-tracing method for large-scale metalenses in multiwavelength imaging system. High Contrast Metastructures XIV. 2025; pp 42–48.
- (8) Shi, J.; Li, D.; Zhang, J.; Wei, X.; Nie, Y. Unified ray-wave model for end-to-end imaging in refractive–diffractive hybrid optics. *Optics Express* **2026**, *34*, 2296–2310.

- (9) Ellepola, K. H.; Rajapaksha, T. D.; Remley, E. E.; Nguyen, M. L. P.; Macdonnell, D. G.; Leckey, J. P.; Vinh, N. Q. Monte Carlo ray-tracing simulations for diffractive optics. *Optics Express* **2026**, *34*, 4465–4480.
- (10) Tokdar, S. T.; Kass, R. E. Importance sampling: a review. *WIREs Computational Statistics* **2010**, *2*, 54–60.
- (11) Jang, E.; Gu, S.; Poole, B. Categorical Reparameterization with Gumbel-Softmax. 2017; <http://arxiv.org/abs/1611.01144>.
- (12) Yin, P.; Lyu, J.; Zhang, S.; Osher, S.; Qi, Y.; Xin, J. Understanding Straight-Through Estimator in Training Activation Quantized Neural Nets. 2019; <http://arxiv.org/abs/1903.05662>, arXiv:1903.05662 [cs.LG].
- (13) Yang, X.; Fu, Q.; Heidrich, W. Curriculum learning for ab initio deep learned refractive optics. *Nature Communications* **2024**, *15*, 6572.